$$f(n,m)=\sum_{i=n}^{?}\sum_{j=m}^{?}(-1)^{i+j}(i,n)(j,m)g(i,j)\iff g(n,m)=\sum_{i=n}^{?}\sum_{j=m}^{?}(-1)^{i+j}(i,n)(j,m)f(i,j)$$